

SEMICONDUCTOR PHYSICS

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SEMI-CONDUCTOR

INSULATORS < Conductivity < CONDUCTORS

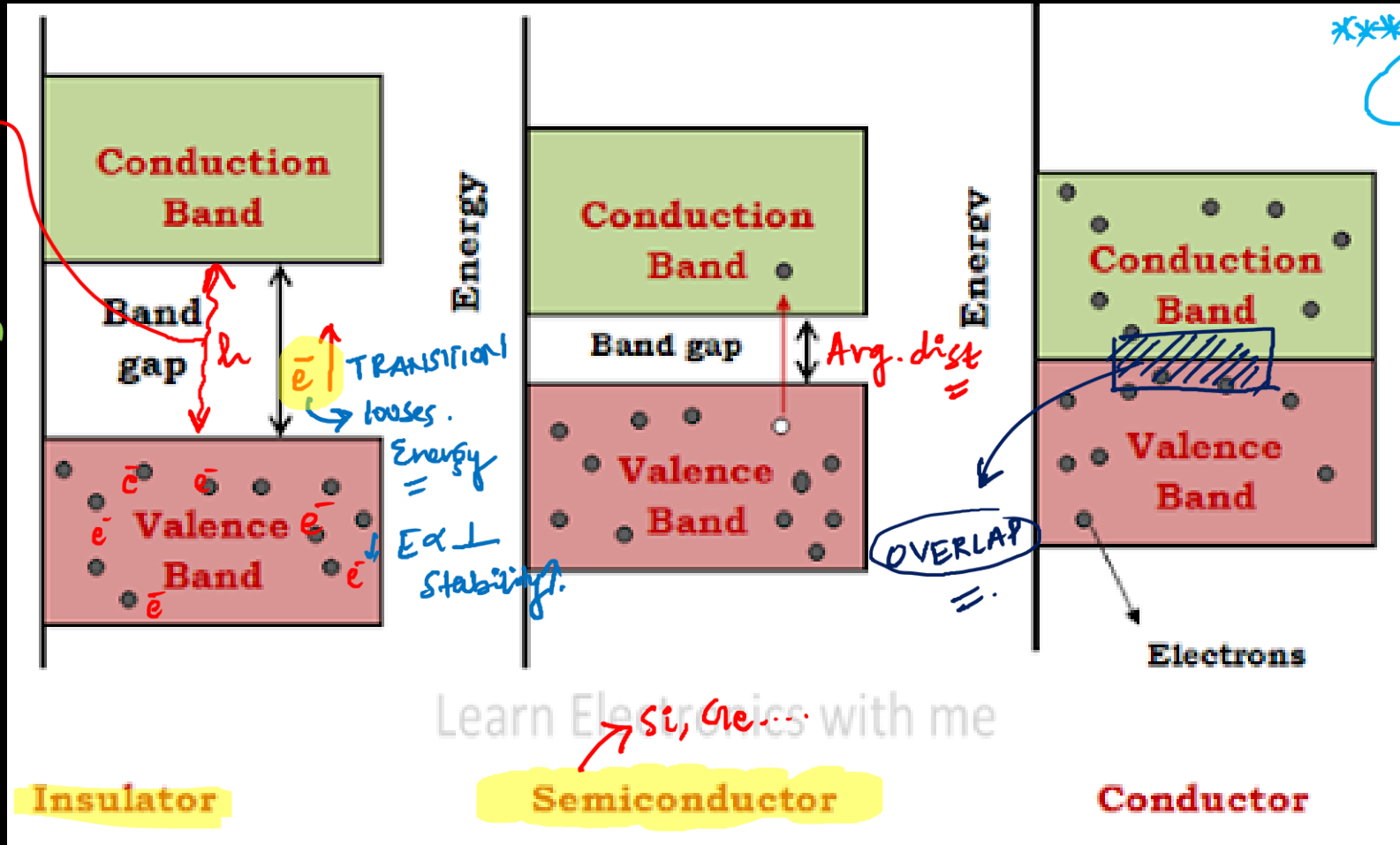
∴ This h is a distance, is a Material dependent property !!

if $h \uparrow$ PE $\uparrow$ TE $\uparrow$ (E $\uparrow$, stability $\downarrow$)

*** dist $\uparrow$ E $\uparrow$ Cond. $\downarrow$

$$E_g = \frac{1.38 \times 10^{-24} \frac{J}{K} \times 300K}{1.6 \times 10^{-19}}$$

$$E_g = 2.5875 \times 10^{-3} \approx 25.87 mV \approx \boxed{26 mV}$$



Learn Electronics with me

Si, Ge...

Insulator

Semiconductor

Conductor

DRIFT VELOCITY = $\frac{1}{2} m v_d^2$

distance $\uparrow$ TE = PE + KE $\uparrow$ mgh
The Energy required to Make a transition from VB $\rightarrow$ CB is w/as BGEL.
[Band Gap Energy level]

It is directly proportional to $\frac{K_B T}{2}$

where $K_B \rightarrow$ Boltzman Constant, $T =$ Absolute Temp. = $27^\circ C = 273 + 27^\circ C = 300K$.
 $= 1.38 \times 10^{-24} J/K$ $2 -$ charge of $e^- = 1.6 \times 10^{-19} C$

CONDUCTIVITY :- distance $\rightarrow$ Insulators $>$ Semi-conductor $>$ Conductors
(h)
Conductivity $\rightarrow$ Conductors $>$ Semi-conductors $>$ Insulators
(σ)

Why Semi-Conductors ?

- $\rightarrow$ $\circ$ EASIER CONTROLLABILITY. (Readily Available).
- $\rightarrow$ $\circ$ LATTICE Structure.
- $\rightarrow$ $\circ$ Thermally generated current.
- $\rightarrow$ $\circ$ Cost Effective.
- $\rightarrow$ $\circ$ Abundance In Availability.

TYPES OF SEMI-CONDUCTORS

INTRINSIC

→ pure form / Natural

→ These are NOT USED
in VLSI Industry

Mostly = ~~X~~

∴ Because they don't
have good conductivity.

ELEMENTAL

→ **Si, Ge**

COMPOUND

**GaAs
InP**

EXTRINSIC

→ Intrinsic + DOPING !!

N-TYPE

- Pentavalent
impurities/dopants

- eg:- Sb, As, Bi, P ✓

P-TYPE

Trivalent
impurities/dopants.

eg- Al, B.

Need of doping ?? → To improve the conductivity of the device (MOSFET, BJT, JFET, pn junction)

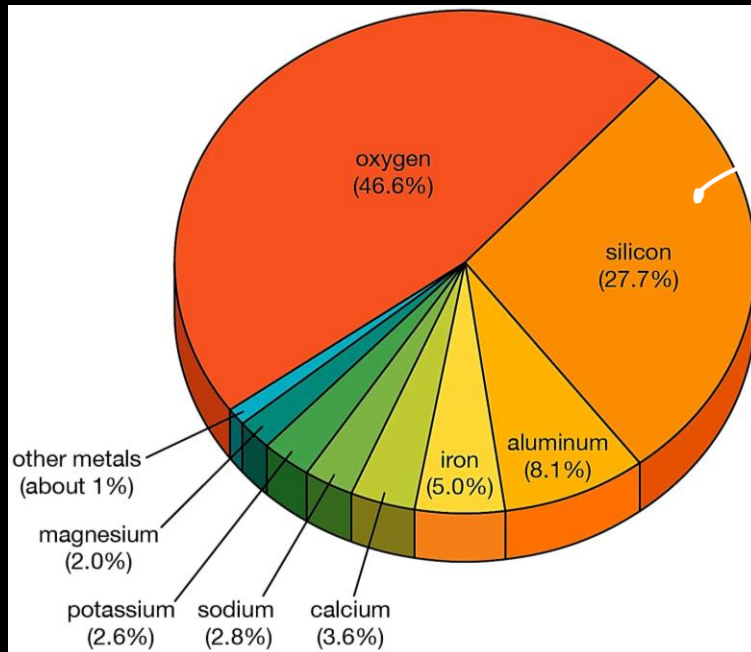
Physical Properties of Silicon

- 1 Its atomic number is 14 with a relative atomic mass of 28.085 u.
- 2 It has a metal-like appearance and is brittle.
- 3 Silicon's density is approximately 2.33 grams per cubic centimeter.
→ Thermally Stable ✓
- 4 Its melting point is 1410 degrees Celsius while its boiling point is 3265 degrees Celsius.
- 5 Silicon isotopes: -28, 29, and 30 are natural, but various artificial isotopes exist that are radioactive.
- 6 Its purest form is an intrinsic semiconductor.



Chemical Properties of Silicon

- 1 Its chemical behavior is purely electropositive. ✓
- 2 Its chemical behavior is very similar to metals.
- 3 It does not easily integrate with oxygen or other related elements because it is solid.
- 4 It is relatively inactive at room temperature.
- 5 At higher temperatures, silicon is highly reactive.
- 6 In a molten state, it forms alloys.



→ 2nd MOST ABUNDANT Mineral found in Earth's crust...

MATERIALS BASED ON CONDUCTIVITY

